

A retrieval-augmented agent bridges virtual screening and automated synthesis

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Abstract

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Abstract: Translating virtually designed drug molecules into executable synthesis protocols is essential for connecting computational discovery with automated synthesis. However, most retrosynthesis methods focus on precursor generation, offer limited route diversity, and lack the detailed reaction conditions required for executable protocols. Here, we present Syn-RRAG, a retrieval-augmented LLM-driven synthesis agent connecting virtual screening with automated synthesis and validation. Syn-RRAG integrates initial retrosynthetic prediction, reaction-precedent retrieval and hierarchical refinement to generate structured protocols with complete experimental parameters. Its local generative model searches over complete reaction hypotheses rather than relying solely on token-level beam search, enabling exploration of feasible routes with distinct reaction classes and bond disconnections. It achieved strong retrosynthesis performance on two patent-derived reaction datasets. Syn-RRAG further generated executable plans for three virtually screened drug candidates, all of which were synthesized and validated on an automated platform. These results establish a workflow connecting virtual molecular design, synthesis planning and experimental validation.

Introduction

Organic synthesis underpins advances in pharmaceuticals¹, materials science² and chemical biology³. A central task in synthetic chemistry is retrosynthetic analysis, whereby a target molecule is recursively decomposed into simpler precursor structures to identify feasible synthetic routes⁴. In recent years, the rapid development of computer-aided drug design and robotic synthesis platforms has made it possible to envision a closed loop for drug discovery between in silico design and wet-lab experimentation^{5,6}. Nevertheless, a critical missing link is the ability to generate diverse and executable synthesis protocols that can be directly implemented on automated systems^{7,8}. Generating such protocols requires not only identifying suitable precursors and alternative routes but also specifying detailed reaction conditions, a process that remains largely dependent on expert knowledge and accumulated experimental experience^{4,9}. Thus, enabling computer-assisted synthesis planning to generate complete, executable and route-diverse experimental protocols represents an important step toward practical AI-assisted synthetic chemistry.

Early computational approaches to retrosynthesis relied largely on rule-based expert systems encoding reaction templates from known transformations¹⁰. Systems such as LHASA⁴ and Chematica^{11,12} demonstrated the feasibility of formalizing expert knowledge but were limited in scalability and adaptability by manual rule curation. More recently, data-driven models trained on large reaction datasets have learned transformation patterns and predicted

plausible precursors directly from molecular representations^{13,14}. However, these models often provide limited reaction conditions and experimental parameters and may struggle to generate multiple viable routes for a target molecule, restricting their utility for practical synthesis planning⁷.

More recently, large language models (LLMs) have emerged as a promising approach for chemical reasoning, enabled by textual representations of molecules such as SMILES^{15–18}. LLMs have shown promise in both retrosynthesis planning and reaction condition optimization^{19–21}. However, most existing approaches treat retrosynthesis as a token-generation-based sequence prediction task and produce incomplete reaction hypotheses, often lacking detailed reaction conditions required for laboratory execution^{8,22}.

Here, we present Syn-RRAG, a self-refining, retrieval-augmented agentic framework for executable retrosynthetic planning. At its core, we design a reaction-state beam search strategy for the fine-tuned LLM, which lifts autoregressive beam outputs from token sequences into chemically grounded reaction states. Instead of accepting top-k reactant SMILES according to sequence likelihood, the model generates candidate reactant sets, canonicalizes and deduplicates them at the molecular level, and ranks the resulting reaction hypotheses using chemical validity, elemental consistency, structural compatibility and forward-prediction checks. This shifts retrosynthetic search from local token variation to reaction-level exploration, allowing Syn-RRAG to retain hypotheses corresponding to distinct reaction classes and bond-disconnection strategies. Our reaction retrieval-augmented generation (RRAG) strategy then maps these reaction states into a learned representation space and retrieves chemically similar precedents from a large patent-derived database. A hierarchical LLM workflow evaluates each hypothesis against these precedents, refines its reaction components and infers missing experimental parameters, producing structured protocols executable on robotic platforms. By coupling reaction-state search with symbolic chemical constraints and precedent-grounded refinement, Syn-RRAG expands route diversity while filtering structurally implausible predictions, thereby improving the generation of feasible experimental protocols. Syn-RRAG achieves strong retrosynthesis accuracy on USPTO-50k and our WIPO patent-derived dataset. Protocols generated for three virtually screened drug molecules were successfully validated on an automated synthesis platform.

Results

Architecture of the retrieval-guided retrosynthesis framework

To generate experimentally executable retrosynthetic plans, we developed a self-refining retrieval-augmented framework that progressively transforms a target molecule into a structured synthesis protocol through an iterative workflow of generation, retrieval, and knowledge-grounded refinement (Fig. 1).

In the generation stage, a supervised fine-tuned (SFT) Llama 3 8B model²³ performs reaction-state beam search by treating beam outputs as candidate reaction-states to be chemically evaluated, rather than directly accepting the top-k reactant SMILES produced by token-level beam search. Given the SMILES representation of the target product, the model first predicts candidate reactants corresponding to plausible retrosynthetic disconnections. The resulting reactant SMILES are canonicalized using RDKit, invalid outputs are discarded and duplicate candidates are merged. The remaining candidates are chemically evaluated and ranked using structural, compositional and forward-prediction criteria. This procedure converts raw beam outputs into a chemically ranked pool of testable reaction hypotheses, enabling the top-ranked candidates to represent distinct disconnection strategies rather than primarily local variations of the same molecular string. Conditioned on both the product and each retained reaction-state, the

model subsequently generates associated reaction conditions, including reagents and solvents. This reaction-state beam search, followed by conditional generation of reaction conditions, captures the dependency between precursor structures and reaction environments and produces structured reaction hypotheses comprising both reactants and initial reaction conditions.

In the retrieval stage, to enable reaction-level knowledge integration, we propose a reaction retrieval-augmented generation strategy, termed RRAG, in which the generated reaction hypothesis is embedded into a learned reaction representation space to enable similarity search against a large-scale, experimentally detailed reaction database extracted from WIPO patent literature. Reactions are encoded using RXNGraphormer^{22,24}, a pre-trained hybrid graph neural network (GNN)-Transformer model that captures both structural and contextual features of chemical transformations. The resulting embedding is used to retrieve the top-10 chemically similar reactions. To improve retrieval relevance, candidate precedents were re-ranked by Gemini-2.5-Flash²⁵ with respect to the predicted transformation, using chemical relevance and mechanistic plausibility as ranking criteria under long-context retrieval settings.

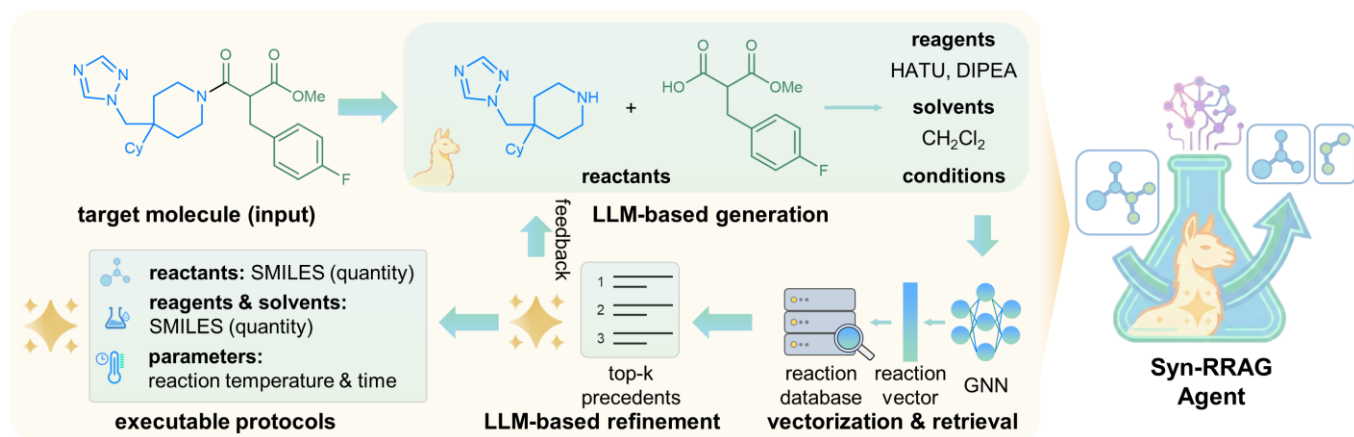


Fig. 1 | Overview of the self-refining retrieval-augmented retrosynthesis framework for executable synthesis planning. Given a target molecule represented as a SMILES string, a fine-tuned LLM first performs reaction-state beam search to generate chemically grounded reaction hypotheses, including candidate reactants and associated reaction conditions (reagents and solvents). The predicted reaction is then embedded into a learned reaction representation space using the RXNGraphormer reaction encoder, enabling similarity-based retrieval of chemically related reactions from a large-scale patent-derived database. Retrieved reactions are re-ranked by a fast, cost-efficient LLM according to chemical relevance and mechanistic plausibility, and the top-ranked precedents are selected as empirical references. A reasoning-optimized LLM performs knowledge-grounded reasoning over these precedents to validate and refine the initial reaction hypothesis and infer missing experimental parameters, including reagent quantities, temperature and reaction time. This generate-retrieve-refine process forms a self-refining workflow that grounds retrosynthetic predictions in experimentally reported chemistry and produces structured, executable synthesis protocols. The framework is implemented as an LLM-assisted synthesis system termed the Syn-RRAG agent.

In the refinement stage, Gemini-2.5-Pro²⁵ evaluates the initial reaction hypothesis against the top-3 re-ranked precedents, revises unsupported components and completes the experimental details required for an executable protocol. Within a predefined protocol schema, the model compares the proposed transformation against related

reported reactions and adjusts reactants, solvents or reagents that conflict with strongly supported precedent patterns. Because numerical parameters such as reagent quantities, temperature and reaction time are insufficiently constrained by supervised fine-tuning alone and are particularly susceptible to unsupported generation, they are inferred with reference to retrieved experimental precedents rather than generated solely from the model’s parametric knowledge. The model then determines these parameters and resolves stoichiometric dependencies to produce an internally consistent protocol. Gemini-2.5-Flash and Gemini-2.5-Pro serve as replaceable reasoning engines for re-ranking and refinement, respectively, and can be substituted with more advanced models.

The full workflow is implemented as an AI-assisted synthesis system termed the Syn-RRAG agentic framework. By coupling generative retrosynthesis, reaction-level similarity retrieval and reasoning-based refinement, the framework grounds predictions in experimentally observed chemistry and enables the generation of structured synthesis protocols suitable for laboratory execution.

Construction of the WIPO-2M reaction dataset

Existing public datasets, such as USPTO-50k, primarily focus on reaction transformations and typically lack experimentally relevant parameters, including reagent quantities, reaction temperature and reaction time. The absence of such information limits their utility for training models that aim to produce executable synthesis protocols. To enable the development of models capable of generating experimentally executable synthesis protocols, we constructed a large-scale reaction dataset containing detailed experimental information extracted from patent literature.

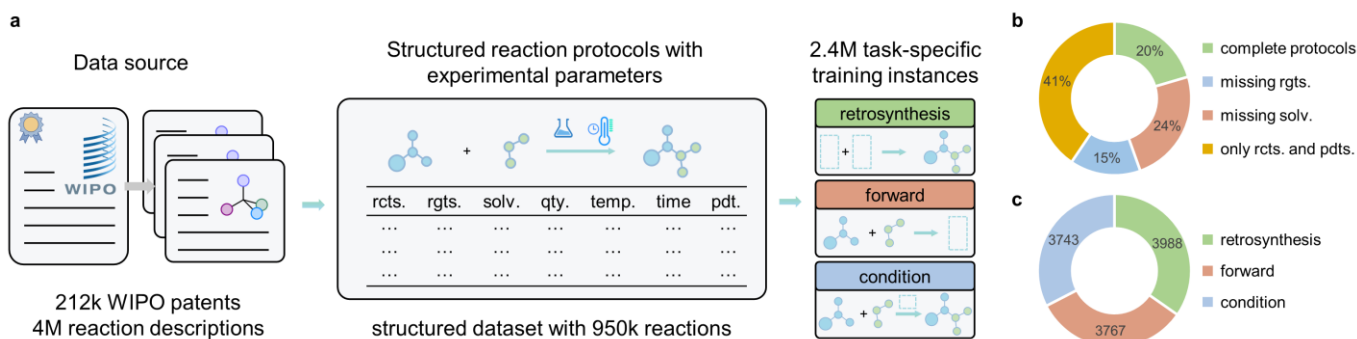


Fig. 2 | Construction and characteristics of the WIPO-2M reaction dataset. **a**, Construction of the dataset from WIPO patent literature. Reaction descriptions are extracted and converted into structured reaction protocols containing reactants, reagents, solvents and experimental parameters, including quantities, temperature and reaction time. The resulting dataset comprises around 950,000 structured reactions and 2.4 million task-specific training instances for retrosynthesis, forward reaction prediction and condition prediction. **b**, Coverage of experimental parameters in the dataset, showing the proportion of reactions with complete protocols, missing reagents, missing solvents and reactants/products only information. **c**, Number of task-specific templates constructed from the WIPO-2M dataset for retrosynthesis, forward reaction prediction and reaction condition prediction.

We curated WIPO patent documents into a structured reaction dataset (Fig. 2a). Approximately 212,000 organic synthesis patents yielded around 4 million reaction-description paragraphs, which were processed using rule-based methods and LLM-assisted parsing, followed by standardization and SMILES mapping. The resulting dataset

contains approximately 950,000 reaction protocols with identified reactants and products; curation details are provided in section 1 of the Supplementary Information.

Importantly, a substantial fraction of the reactions includes experimentally relevant parameters. Approximately 20% of the reactions contain complete experimental information, while 24% lack information about additives and 15% lack solvent annotations (Fig. 2b). Despite partial incompleteness, the dataset provides a significantly richer representation of reactions with experimental details than existing public resources.

Based on this structured reaction dataset, we further constructed 2.4 million task-specific training instances using multiple input–output templates for retrosynthesis prediction, forward reaction prediction and reaction condition prediction (Fig. 2c). This dataset, which we refer to as WIPO-2M, enables the training and evaluation of models that jointly predict reaction precursors, conditions and operational parameters, supporting the generation of executable synthesis protocols.

Evaluation of retrosynthesis prediction and executable protocol generation

Retrosynthesis prediction is the starting point and core step of our synthesis planning pipeline, and it is also a widely benchmarked task^{26–33}. We therefore first evaluated the retrosynthesis performance of the generative backbone within the Syn-RRAG agentic framework on established benchmarks.

To compare our framework with existing retrosynthesis models in a widely used setting, we evaluated the fine-tuned Llama 3 8B model on USPTO-50k³⁴. Following prior non-LLM studies, reaction type labels were included in the input prompts. The model was jointly supervised fine-tuned on forward and retrosynthetic prediction tasks, allowing the same transformation to be learned from both product-to-reactant and reactant-to-product directions and thereby encouraging a more chemically consistent representation of reaction mappings. We selected representative baselines spanning major retrosynthesis paradigms^{26–33}, including template-free²⁹, graph-edit³¹ and graph-to-graph²⁷ methods, together with recent LLM-based approaches^{32,33}. In retrosynthesis performance, Syn-RRAG-Llama-8B achieved top-1, top-3, top-5 and top-10 accuracies of 65.9%, 87.0%, 91.7% and 94.5%, respectively. Encouragingly, it achieved the highest top-1 and top-3 accuracies among all models included in the comparison, while remaining competitive at higher top-n thresholds (Table 1).

Table 1 Comparison of non-LLM and LLM-based models for single-step retrosynthesis on USPTO-50k.

	Models	Top-1	Top-3	Top-5	Top-10
Non-LLM	SCROP ²⁹	59.0	74.8	78.1	81.1
	MEGAN ³¹	60.7	82.0	87.5	91.6
	GraphRetro ²⁸	63.9	81.5	85.2	88.1
	G2G ²⁷	61.0	81.3	86.0	88.7
	LocalRetro ³⁰	63.9	86.8	92.4	96.3
	GLN ²⁶	64.2	79.1	85.2	90.0
LLM-based	ChemDual ³²	50.0	67.7	70.5	78.3
	RetroDFM-R ³³	59.0	-	-	-
	Syn-RRAG-Llama-8B	65.9	87.0	91.7	94.5

We next evaluated Syn-RRAG-Llama-8B on the larger and more diverse WIPO dataset. The model was supervised fine-tuned jointly for retrosynthesis, forward reaction prediction and condition prediction to exploit their chemical coupling. During synthesis protocol generation, the resulting forward prediction capability is also used as a consistency check to assess whether the proposed reactants are likely to produce the target molecule. Products were split between training and test sets by SMILES, ensuring that test products were strictly held out. On this test set, retrosynthesis top-1, top-3, top-5 and top-10 accuracies were 44.5%, 65.1%, 70.8% and 75.9%, respectively. Top-1 accuracies were 75.4% for forward product prediction, 45.8% for solvent prediction and 50.2% for reagent prediction (Supplementary Table 3). Retrosynthesis-only training reduced accuracy to 41.9%, 61.1%, 67.0% and 71.6% at the corresponding thresholds, showing that forward and condition prediction provide complementary supervision. These results support multi-task fine-tuning as the generative backbone of Syn-RRAG for single-step retrosynthesis.

We next evaluated the complete Syn-RRAG framework on synthesis protocol generation for 150 unseen target molecules randomly sampled from the WIPO-2M test set using the full generate–retrieve–refine pipeline. To assess the chemical correctness and experimental feasibility of the generated protocols, we adopted a three-level evaluation framework that also served as a guide to improve the success rate of subsequent automated experimental synthesis validation.

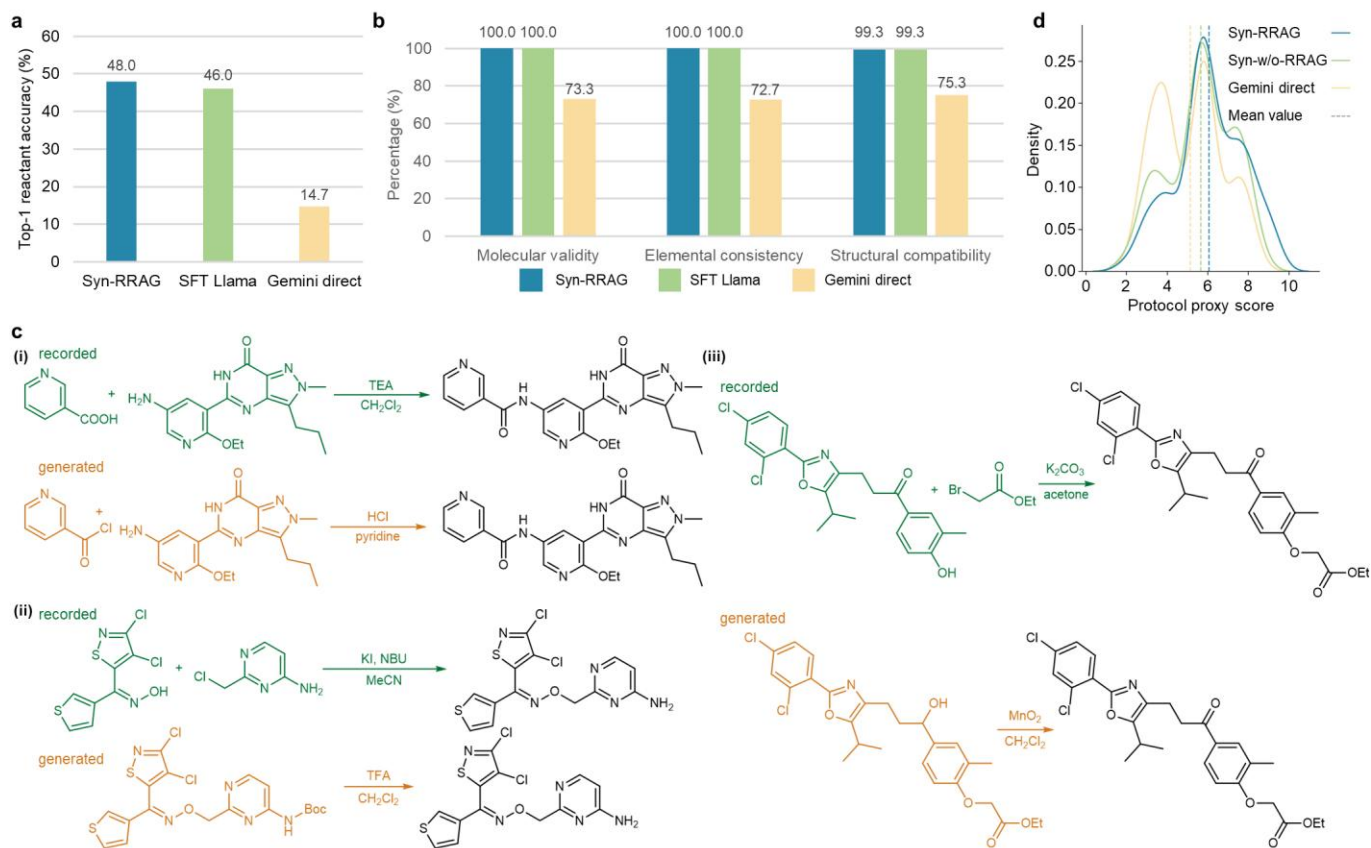


Fig. 3| Multi-level evaluation of synthesis protocol generation. a, Reactant prediction accuracy. **b,** Rule-based structural validation. **c,** Representative examples of chemically plausible synthesis plans generated by Syn-RRAG (yellow) that differ from the recorded reference routes (green) but yield the same target products. **d,** Protocol plausibility assessed by independent LLM evaluator.

The framework generated 150 structured protocols, all convertible into machine-readable records. The top-1 reactant prediction accuracy was 48.0% (Fig. 3a). By comparison, the fine-tuned Llama model without retrieval and refinement achieved 46.0%, and direct generation by Gemini-2.5-Pro (without the fine-tuned retrosynthesis backbone) achieved only 14.7%. These results indicate that retrieval and refinement improve reactant prediction, and that task-specific fine-tuning remains essential beyond general pre-training. Higher top-n accuracies are reported in Supplementary Table 4.

Beyond exact reactant matching, we applied rule-based criteria to evaluate top-1 predictions in terms of molecular validity (RDKit parsability), elemental consistency (all product elements accounted for by the reactants) and structural compatibility (subgraph matching), with further details provided in section 3 of the Supplementary Information. Syn-RRAG achieved 100.0% molecular validity, 100.0% elemental consistency and 99.3% structural compatibility, similar to the fine-tuned model alone (Fig. 3b). These results demonstrate the strong performance of the fine-tuned model in retrosynthetic reactant prediction. Moreover, predictions that do not exactly match the recorded reactants but satisfy elemental and structural constraints may represent chemically feasible alternative routes. By contrast, direct generation by Gemini-2.5-Pro achieved only 73.3% molecular validity, 72.7% elemental consistency and 75.3% structural compatibility, highlighting the susceptibility of general-purpose LLMs to hallucinations in specialized scientific tasks.

Representative non-matching but chemically plausible predictions are shown in Fig. 3c. In panel (i), both the recorded and Syn-RRAG-generated routes form the target through amide-bond formation, using a carboxylic acid and an amine in the recorded route and an acyl chloride and an amine in the generated route. In panel (ii), the recorded route forms an oxime ether through *O*-alkylation of an oxime with a chloromethyl-substituted aminopyrimidine, whereas Syn-RRAG predicts Boc deprotection of an amine, a common and important operation in multistep synthesis³⁵. In panel (iii), the recorded route forms an ether through Williamson-type *O*-alkylation of a phenol with an α -bromo ester, whereas the generated route produces the same target through oxidation of an alcohol to a ketone. These examples demonstrate that Syn-RRAG can propose chemically plausible routes involving reaction types and disconnection strategies distinct from those in the recorded reactions. Consequently, evaluation based solely on exact matching to a single recorded reaction may underestimate its performance. Notably, for all three examples, the recorded reactants were recovered among the top-10 predictions generated by Syn-RRAG.

This route diversity may arise from the combined effects of reaction-state beam search, symbolic chemical verification and training-data coverage. Conventional sequence-based retrosynthesis models typically return candidates directly from token-level beam search^{22,29}, whose outputs often concentrate around a dominant high-probability mode and differ mainly through local token variations; such local variations often fail to construct distinct bond-disconnection strategies and may instead introduce chemically implausible atom- or group-level perturbations. Syn-RRAG instead performs reaction-state beam search by treating beam outputs as complete reaction states: candidate reactants are generated, canonicalized, deduplicated and ranked at the reaction level rather than solely by sequence likelihood. This design allows the search to retain hypotheses involving distinct reaction classes and disconnection strategies. A symbolic chemical verifier further evaluates each hypothesis for molecular validity, elemental consistency and structural compatibility with the target product. By filtering chemically inconsistent candidates without requiring an exact match to a single recorded reaction, the verifier makes broader reaction hypothesis exploration more reliable while preserving feasible alternative routes. In addition, supervised fine-tuning on the large and reaction-diverse WIPO-2M dataset exposes the model to a broad range of transformations and

alternative synthetic strategies. Together, reaction-state beam search, chemical verification and diverse reaction supervision may enable Syn-RRAG to generate multiple chemically plausible routes spanning distinct reaction classes.

As a complementary assessment, we used an independent LLM (GPT-5-pro) to evaluate full-protocol plausibility. This evaluator was not involved in generation or refinement. It assigned each protocol a score from 0 to 10 based on four aspects: appropriateness of temperature/time, plausibility of quantities/stoichiometry, solvent compatibility/concentration, and overall practicality and safety (details can be found in section 7 of Supplementary Information). This scoring can also serve as a pre-experimental filter: low-scoring protocols would be candidates for revision before wet-lab testing, potentially improving the success rate of subsequent experimental validation. Across the 150 test cases, Syn-RRAG achieved a mean protocol proxy score of 6.06, compared with 5.65 for Syn-w/o-RRAG, the variant without retrieval, and 5.12 for direct generation by Gemini-2.5-Pro. The score distributions (Fig. 3d) show that Syn-RRAG generates more high-scoring protocols and achieves the highest overall mean. The consistent ranking across these settings indicates that the retrosynthesis-specialized backbone substantially improves protocol quality, while reaction retrieval and refinement provide a further performance gain. Together, these three levels of evaluation provide complementary evidence on protocol quality, from structural correctness to experimental plausibility, while also directly supporting the design of successful wet-lab experiments.

From in silico molecular design to automated synthesis

With the agentic framework established, we next asked whether it could bridge computational molecular discovery with executable laboratory synthesis. To this end, we applied the tool to three representative therapeutic targets: the hepatocyte growth factor (HGF) receptor **MET**, the epidermal growth factor receptor (**EGFR**), and interleukin-17A (**IL17A**), all of which are closely associated with cancer progression or inflammation-related diseases (Fig. 4a). Starting from 100,000 compounds from the Enamine Hit Locator Library, we first performed large-scale molecular docking against each target to rapidly search potential binders. The top-ranked candidates were then further filtered using synthetic accessibility scores to prioritize compounds with preferred synthesizability. For each target, the top 100 synthetically accessible candidates were selected for more rigorous computational evaluation by molecular dynamics (MD) simulations and absolute binding free energy (ABFE) calculations. This hierarchical workflow ultimately identified one representative ligand for each target: 2-(*m*-tolyl)quinazolin-4(3*H*)-one (**1c**) for **MET**, *N*-(2,5-dimethylphenyl)quinoline-2-carboxamide (**2c**) for **EGFR**, and (2*S*)-2-[(3-chlorophenyl)carbamoylamino]-3-(1*H*-indol-3-yl)-*N*-methylpropanamide (**3c**) for **IL17A**. Further details of the virtual screening workflow are provided in section 8 of Supplementary Information.

We then used the Syn-RRAG agent to design complete and executable synthesis protocols for these three unseen molecules (Fig. 4b). For the three target ligands, Syn-RRAG generated three chemically distinct and practically feasible synthetic strategies. For **1c**, the agent proposed an iodine-mediated oxidative cyclocondensation between an anthranilamide derivative and an aromatic aldehyde to construct the quinazolin-4(3*H*)-one scaffold. For **2c**, it designed a hexafluorophosphate azabenzotriazole tetramethyl uronium (HATU) mediated amide coupling between a quinoline carboxylic acid and a substituted aniline. For **3c**, it proposed an amine–isocyanate addition to form the substituted urea linkage. These examples cover heterocycle formation, amide-bond formation, and urea-bond formation, respectively, highlighting the ability of the agent to adapt its synthesis planning strategy to different molecular scaffolds and bond-forming requirements.

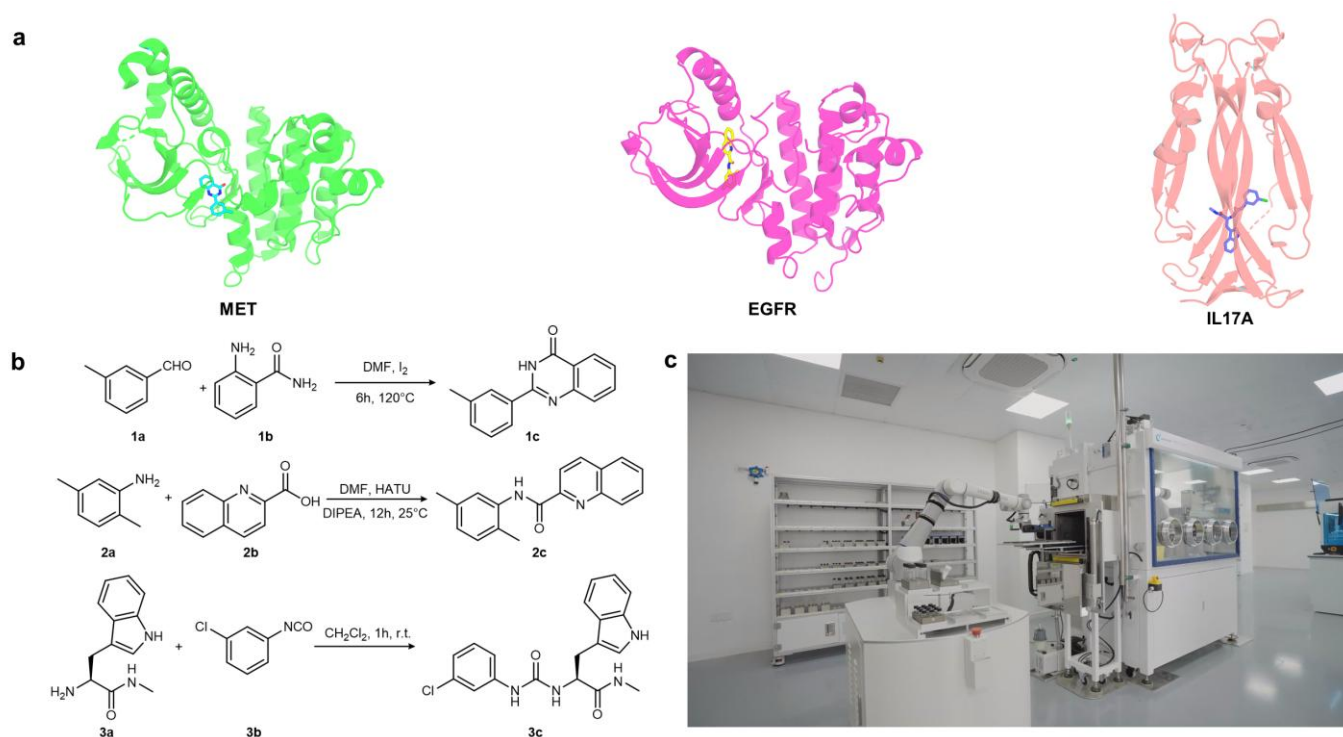


Fig. 4| From in silico molecular design to automated synthesis. **a**, Predicted binding poses of the virtually screened ligands **1c**, **2c** and **3c** within their respective protein targets, **MET**, **EGFR** and **IL17A**. **b**, Executable synthesis routes generated by Syn-RRAG agent for ligands **1c**, **2c** and **3c**, including the reactants, reagents, solvents, reaction times and temperatures. DMF, dimethylformamide; HATU, hexafluorophosphate azabenzotriazole tetramethyl uronium; DIPEA, *N,N*-diisopropylethylamine; r.t., room temperature. **c**, Photograph of the Golab automated synthesis platform used to execute and experimentally validate the generated synthesis protocols.

Beyond route design, Syn-RRAG generated full experimental protocols that included the reactant structures, reagent equivalents, solvent, additives or base, reaction temperature, and reaction time. This level of procedural detail is essential for connecting LLM-based synthesis planning with autonomous experimentation, as automated laboratory platforms require unambiguous and parameterized instructions rather than high-level reaction schemes alone. The detailed synthetic protocols for the three ligands are provided in section 9 of Supplementary Information.

To experimentally validate the generated protocols, we submitted the three complete synthesis plans to our in-house Golab automated synthesis platform (Fig. 4c). The platform integrates robotic operation with automated modules for reaction setup, sampling, concentration, purification, and analytical characterization. All three out-of-sample molecules were successfully synthesized and confirmed as the desired products. The video of the automated experiments is provided as Supplementary Video 1. Beyond validating individual synthesis plans, this application establishes an end-to-end workflow connecting in silico molecular design, executable synthesis planning and automated experimental validation, providing a practical approach to accelerating early-stage drug discovery.

Discussion

In this work, we developed Syn-RRAG, a self-refining, retrieval-augmented agentic framework for generating complete and executable synthesis protocols. Syn-RRAG uses a fine-tuned Llama 3 8B model with reaction-state

beam search to generate diverse retrosynthetic hypotheses, enabling exploration of multiple plausible synthesis routes with distinct reaction classes and bond-disconnection strategies. It then retrieves experimental precedents based on whole-reaction similarity and performs precedent-grounded refinement to determine reaction components and operational parameters. This generate–retrieve–refine architecture grounds generative predictions in reported chemistry and converts retrosynthetic hypotheses into machine-readable experimental protocols. By applying Syn-RRAG to compounds identified through virtual screening and executing the generated protocols on an automated synthesis platform, we demonstrate its ability to bridge in silico molecular discovery and wet-lab validation.

A key resource supporting Syn-RRAG is WIPO-2M, a reaction dataset containing detailed experimental information, constructed from approximately 212,000 WIPO patents and 4 million reaction-description paragraphs. It contains approximately 950,000 structured reactions and 2.4 million task-specific instances for retrosynthesis, forward reaction prediction and condition prediction. By incorporating detailed reaction components and operational parameters, WIPO-2M supports both model training and reaction-precedent retrieval for executable synthesis planning.

Syn-RRAG was evaluated across multiple tasks and levels. On USPTO-50k, Syn-RRAG-Llama-8B achieved top-1 and top-3 retrosynthesis accuracies of 65.9% and 87.0%, respectively, outperforming the representative models included in our comparison. On the WIPO test set with held-out products, it achieved top-1 accuracies of 44.5% for retrosynthesis, 75.4% for forward prediction, 45.8% for solvent prediction and 50.2% for reagent prediction. A three-level evaluation of 150 protocols generated for previously unseen target molecules further assessed reactant accuracy, structural consistency and experimental plausibility, showing that Syn-RRAG generated chemically consistent and experimentally plausible protocols. Analysis of predictions that differed from the recorded reactions further demonstrated its ability to propose multiple chemically reasonable synthesis routes for the same target molecule.

We further applied Syn-RRAG to three candidate molecules identified through virtual screening against the protein targets **MET**, **EGFR** and **IL17A**. The framework generated executable reaction protocols involving oxidative cyclocondensation, amide coupling and urea formation, all of which were successfully executed and validated on the Golab automated synthesis platform. These results establish a practical connection between virtual molecular discovery, synthesis planning and automated experimentation. Broader validation across chemical space and multistep transformations will nevertheless be required. Future integration with automated reaction-mechanism analysis³⁶ could complement precedent-based planning by assessing proposed transformations mechanistically and diagnosing unsuccessful reactions, thereby guiding route and condition refinement. Together, executable planning, automated experimentation and mechanistic feedback could move drug discovery towards a closed loop spanning in silico molecular design, synthesis planning and experimental validation.

Methods

The generate–retrieve–refine architecture

To bridge the execution gap between exact molecular sequence prediction and the natural language generation of an executable synthesis protocol, we define a tripartite probabilistic framework: generate–retrieve–refine. Let P denote the target product, $\mathcal{T}_{\text{core}}$ the stoichiometric skeleton (reactants), C_{prior} the theoretically predicted reaction conditions, and $\mathcal{K}$ the retrieved empirical knowledge. Our system models the joint probability distribution of the final executable protocol, Y_{protocol} , via the following factorization:

$$(Y_{\text{protocol}}, \mathcal{T}_{\text{core}}, C_{\text{prior}}, \mathcal{K} \mid P) = \underbrace{P_{\text{gen}}(\mathcal{T}_{\text{core}}, C_{\text{prior}} \mid P)}_{\text{generation}} \cdot \underbrace{P_{\text{ret}}(\mathcal{K} \mid \mathcal{T}_{\text{core}} \oplus P)}_{\text{retrieval}} \cdot \underbrace{P_{\text{ref}}(Y_{\text{protocol}} \mid P, \mathcal{T}_{\text{core}}, C_{\text{prior}}, \mathcal{K})}_{\text{refinement}}$$

Generation of initial reaction hypotheses through joint multi-task fine-tuning on WIPO-2M

The first module acts as a task-specific predictive engine. Although architecture-agnostic, we implemented this module using an instruction-tuned foundation model (Llama 3 8B). Its primary function is to constrain the permissible chemical state space before procedural protocol generation.

Reaction-state beam search: Operationally, this strategy proceeds in four steps: proposal, lifting, verification and expansion. First, the SFT model generates multiple reactant SMILES candidates using beam decoding. Second, each decoded sequence is parsed by RDKit and lifted into a canonical reactant state; invalid molecules are discarded and duplicate canonical states are merged. Third, each reactant state is scored using chemical validity, elemental consistency, scaffold compatibility, abnormal-fragment penalties and, when enabled, forward-prediction consistency. Finally, the retained reactant states are expanded into complete reaction states by condition generation conditioned on both the product and the reactants.

The model was jointly fine-tuned on the WIPO-2M dataset for retrosynthesis, forward reaction prediction and reaction-condition prediction. Rather than treating these tasks as mutually exclusive pathways, we optimized them simultaneously through joint representation learning. For each reaction instance in WIPO-2M, the model parameters were updated by combining the losses associated with all available reaction entities.

Joint state mapping and dense supervision: Let a reaction instance from the dataset $\mathcal{D}$ be defined by the tuple $(\mathcal{T}_{\text{core}}, P, C)$, denoting the reactants, product, and environmental conditions (reagents/solvents), respectively. Let $m_c \in \{0,1\}$ act as a Boolean availability mask denoting whether empirical conditions are present in the source literature. For each reaction instance, we construct up to three distinct instruction-response pairs:

- **Retrosynthesis** (τ_{retro}): The input is the product ($X_{\text{retro}} = P$), and the target sequence is the reactants ($Y_{\text{retro}} = \mathcal{T}_{\text{core}}$).
- **Forward prediction** (τ_{fwd}): The input conditionally incorporates the reaction conditions. Thus, $X_{\text{fwd}} = \mathcal{T}_{\text{core}} \oplus (m_c \cdot C)$, and the target is the resulting product ($Y_{\text{fwd}} = P$).
- **Condition identification** (τ_{cond}): The input concatenates both the reactants and the product ($X_{\text{cond}} = \mathcal{T}_{\text{core}} \oplus P$), and the target is the environmental conditions ($Y_{\text{cond}} = C$).

Let $l(X, Y)$ denote the standard autoregressive negative log-likelihood for a given prompt X and target sequence Y :

$$l(X, Y) = - \sum_{t=1}^{|Y|} \log P_{\theta}(y_t \mid y_{<t}, X)$$

Because the tasks are optimized jointly, the total multi-task instruction-tuning loss $\mathcal{L}_{\text{SFT}}(\theta)$ for a given reaction tuple is defined as the weighted sum of these concurrent objectives:

$$\mathcal{L}_{\text{SFT}}(\theta) = \mathbb{E}_{(\mathcal{T}_{\text{core}}, P, C) \sim \mathcal{D}} [l(X_{\text{retro}}, Y_{\text{retro}}) + \lambda_{\text{fwd}} l(X_{\text{fwd}}, Y_{\text{fwd}}) + m_c \cdot \lambda_{\text{cond}} l(X_{\text{cond}}, Y_{\text{cond}})]$$

where λ_{fwd} and λ_{cond} serve as regularization weights (both set to 1 in our training) to ensure the auxiliary forward and condition tasks stabilize, rather than overpower, the primary retrosynthetic objective. The availability

mask m_c naturally drops the condition-identification loss when data is sparse, ensuring the model defaults to learning base distributions. This simultaneous summation provides dense supervision, forcing θ to learn a bidirectional, fully constrained representation of chemical reactivity.

Dataset partitioning for multi-task fine-tuning: For multi-task fine-tuning on WIPO-2M, the dataset was partitioned into training, validation and test sets using a 98:1:1 ratio.

Reaction-level precedent retrieval and re-ranking

Standard text-based retrieval fails to capture the strict structural constraints of chemical reactions. We frame the retrieval task as a two-stage process: graph-based embedding space projection followed by listwise semantic reranking. This cascade ensures the system retrieves precedents that share the core reaction topology while providing precise empirical conditions (time, temperature, and experimental procedures).

Structural coarse retrieval: We use the RXNGraphormer encoder to map each chemical transformation, comprising the target product and core reactants, into a dense embedding space. The offline retrieval database contains 164,000 curated reactions drawn exclusively from the WIPO-2M training set, with fully annotated reaction entities (reactants, reagents, solvents and products) and corresponding experimental procedures extracted from patents. The search index is built using Facebook AI Similarity Search (FAISS) to compute Euclidean distances within the RXNGraphormer embedding space. Given the target product molecule and predicted core skeleton reactants, denoted as the core reaction pair $\mathcal{T}_{\text{core}} \oplus P$, we search this offline database to retrieve a coarse set of k candidates $\mathcal{K}_{\text{coarse}} = \{c_1, c_2, \dots, c_k\}$. Let c_i represent a candidate reaction from the patent database $\mathcal{D}_{\text{patent}}$. The set is obtained by minimizing the L_2 distance in the vector space:

$$\mathcal{K}_{\text{coarse}} = \arg \min_{\mathcal{S} \subset \mathcal{D}_{\text{patent}}, |\mathcal{S}|=k} \sum_{c_i \in \mathcal{S}} \|f_{\phi}(c_i) - f_{\phi}(\mathcal{T}_{\text{core}} \oplus P)\|_2^2$$

Listwise semantic reranking: To improve retrieval relevance, the coarse candidate precedents are re-ranked with respect to the predicted target transformation, utilizing chemical relevance and mechanistic plausibility as the primary ranking criteria under long-context retrieval settings. Rather than scoring candidates individually, we utilize a lightweight, high-speed LLM parameterized by ψ (Gemini-2.5-Flash) to perform in-context listwise reranking. We concatenate the entire coarse candidate set into a joint prompt context, prompting the model to evaluate the joint distribution of all candidates simultaneously to directly output the optimal subset $\mathcal{K}_{\text{fine}}$ of size m (where $m = 3$):

$$\mathcal{K}_{\text{fine}} = \arg \max_{\mathcal{S} \subset \mathcal{K}_{\text{coarse}}, |\mathcal{S}|=m} P_{\psi}(\mathcal{S} \mid \mathcal{T}_{\text{core}} \oplus P, c_1 \oplus c_2 \oplus \dots \oplus c_k)$$

This listwise formulation allows the attention mechanism of the reranking model to cross-reference constraints among the candidates themselves. By evaluating the precedents directly against one another, the system effectively filters out lower-relevance textual matches and prioritizes the precedents that offer the most appropriate operational context for the downstream protocol synthesis.

Precedent-guided evaluation and refinement

The final stage employs a cutting-edge generative model $\mathcal{M}_{\text{gen}}$ (Gemini-2.5-Pro) to evaluate and refine the initial reaction hypothesis and complete the experimental details required for an executable synthesis protocol. We formalize this generation as an approximate Maximum a Posteriori (MAP) inference task, where the model acts as a

context-aware Bayesian updater. It is prompted to generate the final procedural tokens, Y_{protocol} , by interpolating between the theoretical inductive prior (C_{prior}) supplied by the local model and the empirical evidence ($\mathcal{K}_{\text{fine}}$) supplied by the retrieval vector space, conditioned on the target molecule P and the core reactants $\mathcal{T}_{\text{core}}$ predicted by the local model. The objective maximizes the conditional log-likelihood over the vocabulary space $\mathcal{V}^*$:

$$Y_{\text{protocol}} = \arg \max_{Y \in \mathcal{V}^*} \sum_{t=1}^{|Y|} \log P_{\text{ref}}(y_t \mid y_{<t}, P, \mathcal{T}_{\text{core}}, C_{\text{prior}}, \mathcal{K}_{\text{fine}})$$

Operationally, this inference is executed through a strictly constrained in-schema Chain-of-Thought (CoT) framework. The prompt supplies the generative agent with the complete context: the local SFT-generated conditions, the top-3 retrieved and reranked experimental precedents, the target product, and the recommended reactants. Prior to drafting the final text, the model is instructed to conduct a comparative analysis of these inputs and explicitly articulate its rationale for optimizing the chemical entities within a predefined JSON schema. By forcing the model to document this comparative reasoning step, the system ensures the final generated protocol is strictly grounded in verified empirical data rather than unverified parametric heuristics.

Rule-based structural verification and iterative feedback

Each generated reaction hypothesis is evaluated using rule-based checks for molecular validity, elemental consistency and structural compatibility. Molecular validity is determined by RDKit parsability, elemental consistency verifies that the elements in the product are represented among the reactants, and structural compatibility is assessed through subgraph matching between reactants and the target product. Failed checks are converted into explicit feedback that guides the model to revise invalid reactants or inconsistent reaction components in the next iteration, thereby improving the chemical consistency of the generated synthesis protocol.

Data availability

The 2.4 million task-specific instruction-tuning instances and the weights of the supervised fine-tuned model are available through GitHub at <https://github.com/SamSamChu/Agentic-Syn-RRAG>.

Code availability

The scripts for model training, evaluation and generation of complete experimental protocols are available through GitHub at <https://github.com/SamSamChu/Agentic-Syn-RRAG>.

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Author contributions

S.Z.: methodology, software, validation, investigation, and writing - original draft. X.C.: software. P.W. and Q.J.: validation and writing - review and editing. M.L., Q.W., and G.Z.: data curation. Y.Q. and F.C.: supervision,

resources, and funding acquisition. L.-C.X.: conceptualization, visualization, project administration, supervision, funding acquisition, writing - original draft, and writing - review and editing.

Competing interests

The authors declare no competing interests.

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